

ABSTRACT

A two-center safety and efficacy clinical study with a novel home-use 1440 nm skin rejuvenating laser: A 3-month evaluation

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Background & Objectives:

An investigational 1440 nm non-ablative fractional diode laser was developed for consumer use in the treatment of facial fine lines and wrinkles, dyschromia, and tactile skin roughness. This study was designed to demonstrate safety and efficacy of the home-use laser device.

Study Design/Materials and Methods

Ninety subjects were enrolled in a prospective clinical study. Each subject was randomly assigned to one of three treatment levels (low, medium, high) ranging from 36 to 130 microbeams/cm² at 5 to 12 mJ/microbeam. Subjects self-treated their entire face daily (Monday through Friday) at the study site for 12 weeks and were followed for 12 weeks after the final treatment. Satisfaction questionnaires were completed 2, 4, and 8 weeks during treatment and at each follow-up visit. Periorbital wrinkles and dyschromia were assessed by a panel of 3 independent, blinded expert graders who scored standardized photographs using the 9-point Fitzpatrick Wrinkle scale and a validated 9-point dyschromia scale, respectively. Tactile roughness was assessed every two weeks by the investigator using a 9-point scale. Tolerability and adverse events were monitored throughout the study.

Results

Eighty-seven subjects completed the study. Statistically significant

improvement ($p < 0.001$) in periorbital wrinkles, dyschromia, and tactile roughness was found for all treatment cohorts 12 weeks post final treatment. Mean periorbital wrinkle scores improved by 0.6, 0.8, and 0.8 points for the low, medium, and high cohorts, respectively. Mean dyschromia scores improved by 0.4, 0.7, and 1.0 points for the low, medium, and high cohorts, respectively. Mean tactile roughness scores improved by 2.4, 2.6, and 2.8 points for the low, medium, and high cohorts, respectively. At least 73.8% of all subjects reported improvement in wrinkles, brown spots, and texture and 77.4% were satisfied with their treatments. The most common adverse device effects consisted of mild erythema, stinging, and warm or burning sensation with 99% of cases resolving within hours after treatment. Treatments were very well tolerated with the mean tolerability score during the first day of treatment being 2.0 (out of 10) which steadily reduced to 0.6 by the last treatment.

Conclusions

The study demonstrated that the home-use 1440 nm non-

ablative fractional laser resulted in statistically significant and clinically meaningful improvement in periorbital wrinkles, dyschromia, and tactile roughness. Treatment was extremely well tolerated and had minimal risk to participants. In summary, this technology is a non-invasive option for consumers interested in treating photodamaged skin at home.

Background

In the past decade, fractional non-ablative laser treatments have emerged as an effective alternate to ablative lasers for treating multiple signs of aging. Fewer adverse events and faster recovery times are among its advantages. Only recently has this technology emerged into the home-use market. A new 1440 nm non-ablative, fractional laser was developed for consumer use to treat wrinkles, dyschromia, and tactile roughness. This device features several treatment levels for user comfort. Safety mechanisms are built into the device to prevent overtreatment when the device is stationary and to prevent the device from firing when not in contact with the skin.

A previous investigation explored the wound morphology and healing response of this home-use non-ablative, fractional laser. The histological images produced (Figure 1) show thermal injuries that have a depth and healing profile similar to that of professional non-ablative, fractional devices.

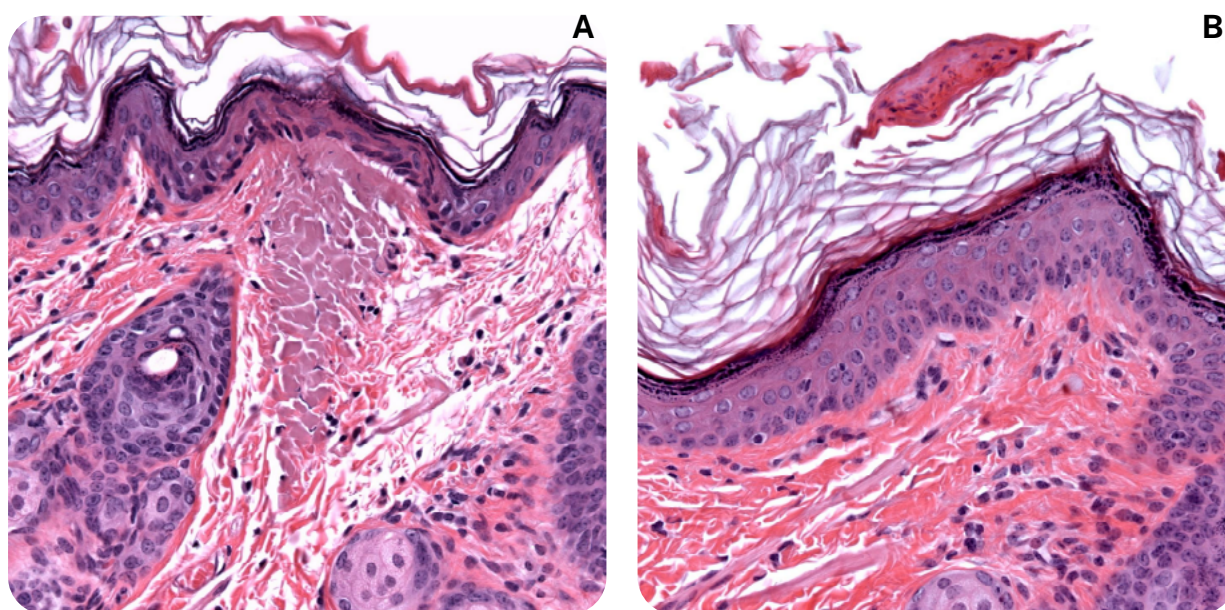


Figure 1: A) A microthermal zone produced by the 1440 nm non-ablative fractional home-use laser immediately after treatment (250 μ m deep x 90 μ m wide)¹. B) Microscopic epidermal necrotic debris found 5 days after treatment. The epidermis is fully intact.

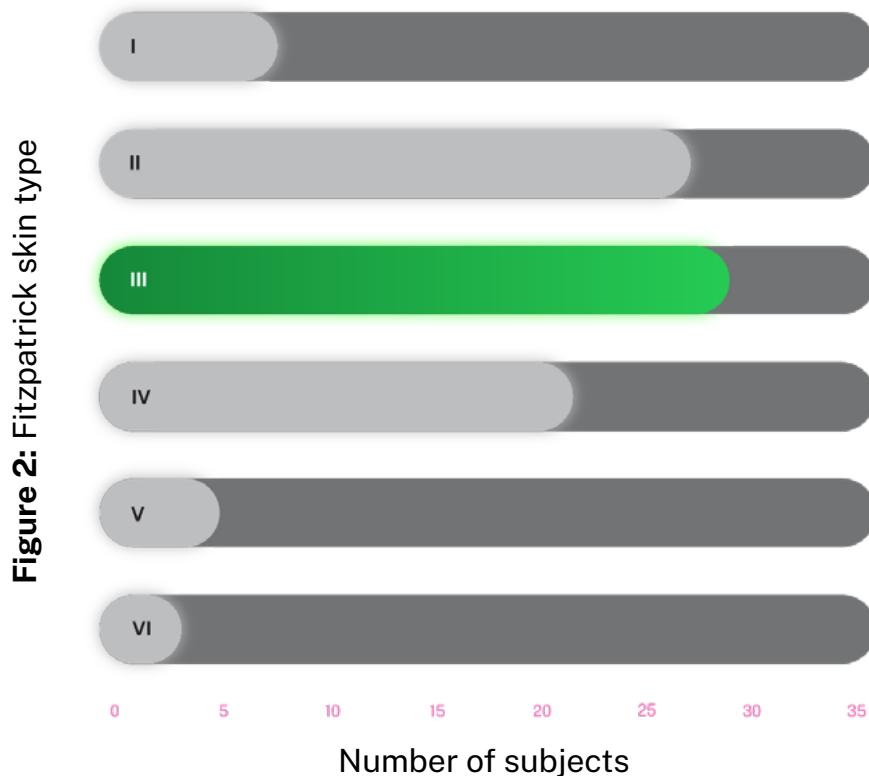
¹Lee, J., Biesman, B., Ostrowski, R., Histopathology of a home-use 1440 nm diode laser for skin rejuvenation. 71st American Academy of Dermatology Summer Meeting, 2013 July 30 – Aug 3, New York, Poster A7185

Objective

This clinical study was conducted to determine the safety and efficacy of a home-use laser for treating facial photodamage including periorbital wrinkles, dyschromia, and textural irregularities such as tactile skin roughness.

Population

- 90 subjects enrolled into a randomized open-label study
- 87 female, 3 male
- Age range was 32 – 70 years (mean 53.4 ± 8.0)
- 87 White, 3 Black/African American
- 19 Hispanic/Latino
- All had periorbital wrinkles, dyschromia, and tactile skin roughness on both sides of the face
- All Fitzpatrick skin types enrolled (Figure 2)



Exclusion

- Pregnancy or breastfeeding
- Known allergy or sensitivity to topicals or study supplies provided in the study
- Medical or cosmetic surgery on the face
- Facial cosmetic procedures within 6 months of the baseline visit
- Dermal filler injections on the face
- Use of topical or systemic steroids, topical retinoids, chemical exfoliants, depigmentation products within 4 weeks of the baseline visit
- Use of systemic retinoids within 1 year of the baseline visit
- Infections, skin disorders, or visible scars on the face
- Tanned or sunburned skin on the face
- Smoking within 1 year of the baseline visit

Treatment/Procedures

- Subjects randomly assigned to one of three treatment cohorts
 - Energy 5-12 mJ
 - Density 36-130 microbeams/cm²
- Subjects self treated their entire face (Figure 3) at the study site 5 days per week for 12 weeks and returned for follow-up visits at 1 day, 2, 4, 8, and 12 weeks post final treatment
- At each treatment, subjects rated their pain from treatment using a visual analogue scale, where 0 is no pain and 10 is extreme pain
- Adverse events were monitored throughout the study
- Satisfaction questionnaires were completed 2, 4, and 8 weeks during treatment and at each follow-up visit
- Photos of subjects were taken every two weeks during treatment and at each follow-up visit using the Canfield VISIA-CR system

Evaluations

- Three blinded expert physicians evaluated periorbital wrinkles and dyschromia using standardized photos at baseline, 4 weeks post and 12 weeks post final treatment
 - Before and after photos were randomized
 - The 9-point Fitzpatrick Wrinkle Assessment Scale was used to assess periorbital wrinkles
 - The 9-point Dyschromia Assessment Scale (validated) was used to assess dyschromia
- The investigators evaluated tactile roughness at baseline, every 2 weeks during treatment, and at each follow-up visit using the 9-point Tactile Roughness Assessment Scale (back of the hand assessment)



Figure 3: Treatment areas on the face shaded in green.

RESULTS AND CONCLUSIONS 12 WEEKS POST FINAL TREATMENT



Baseline (left) and 12 weeks post final treatment (right)



Baseline (left) and 12 weeks post final treatment (right)



Baseline (left) and 12 weeks post final treatment (right)

Study photos were taken at KGL, Inc and Baumann Cosmetic and Research Institute

Evaluation Results for Periorbital Wrinkles, Dyschromia, and Tactile Roughness (Table 1)

- Significant improvement was observed for all treatment cohorts
- Mean improvement in scores for all subjects were 0.7 ± 0.6 , 0.7 ± 0.6 , and 2.6 ± 1.2 for periorbital wrinkles, dyschromia, and tactile roughness, respectively
- Higher cohorts displayed higher mean improvement across all indications studied

Indication	Low cohort	Medium cohort	High cohort
Periorbital wrinkles	$0.6^* \pm 0.6$	$0.8^* \pm 0.6$	$0.8^* \pm 0.7$
Dyschromia	$0.4^* \pm 0.7$	$0.7^* \pm 0.6$	$1.0^* \pm 0.5$
Tactile roughness	$2.4^* \pm 1.3$	$2.6^* \pm 1.5$	$2.8^* \pm 0.8$

Table 1: Mean ($\pm$ SD) reduction in scores (by cohort) evaluated by blinded expert panel (wrinkles and dyschromia) and investigator (roughness) at 12 weeks post final treatment.

* Statistically significant ($p < 0.001$).

Subjective questionnaire (Table 2)

- At least 73.8% of all subjects reported improvement in wrinkles, brown spots/ areas, and texture
- 77.4% of all subjects were satisfied with their treatment

Improvement in... (% of subjects)	Low cohort	Medium cohort	High cohort
Fine lines & wrinkles (around eyes)	75.9%	71.4%	77.8%
Fine lines & wrinkles (around other areas)	72.4%	75.0%	77.8%
Brown areas/spots	65.5%	75.0%	81.5%
Skin texture	75.9%	78.6%	88.9%
Overall satisfaction	69.0%	85.7%	77.8%

Table 2: Percent of subjects reporting results (by cohort) at 12 weeks post final treatment

Side effects and tolerability (Table 3)

- Treatments were very well tolerated with mean pain scores for all treatment levels starting low (0.5-2.6 out of 10) and decreasing steadily throughout study
- No severe or serious adverse device effects were reported
- Most common device related incidents were erythema, stinging/prickling, and warm/burning sensation
- Most (99%) events were reported as trace to mild in severity and resolved within hours after treatment

Treatment week	Low cohort	Medium cohort	High cohort
Week 1	0.5 ± 0.5	1.6 ± 1.5	2.6 ± 1.8
Week 12	0.3 ± 0.3	0.6 ± 0.7	1.2 ± 1.2

Table 3: Mean (±SD) pain score (0-10 scale) by cohort

Conclusions

This 1440 nm non-ablative, fractional laser is a safe and effective option for at-home treatment of periorbital wrinkles, dyschromia, and tactile roughness.

- Clinically significant improvement was observed with higher efficacy apparent at higher levels
- Participants were satisfied with their treatment and saw improvement
- Treatments were extremely well tolerated with trace to mild side effects and minimal to no downtime